B-cell epitope prediction from antigen-only dynamics: project log

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Living record. Read top to bottom to get up to speed.

Running record of the project: predict B-cell epitope residues from *antigen-only* information, using alanine-scan $\Delta\Delta G_{\text{bind}}$ as residue-level binding-importance labels and asking whether antigen dynamics / solvation / energetics carry predictive signal *beyond static surface exposure*. Read in order; each section is one iteration — what we tried, what we found, what it changed. Not a formal manuscript. Every figure is regenerated by a named script (recorded in `figures/FIGURES.md`); every number traces to a file or command; unconfirmed values are flagged `[verify]`.

1 Motivation and the gap

We want to predict which residues of an antigen are B-cell epitope residues, from the antigen structure alone, in cases where the antibody is unknown. Doing this rigorously requires residue-level labels for "binding importance." We take these from **alanine scanning**: for residue i, $\Delta\Delta G^{(i)}_{\text{bind}} = \Delta G^{\text{mut},i}_{\text{bind}} - \Delta G^{\text{WT}}_{\text{bind}}$, with large positive values flagging residues whose wild-type identity stabilizes the interface. We treat this as a residue-level epitope score, $S^{\text{epitope}}_i \approx \Delta\Delta G^{(i)}_{\text{bind}}$. (Full derivation and the alchemical / StaB-ddG label tiers: `../../tcr/docs/b_cell_eptiope_prediction_may23_update_final.pdf`

The gap. Most B-cell epitope predictors label residues by *contact distance* to a bound antibody. That conflates "close" with "important": a residue can sit at the interface yet contribute little energetically, or be energetically critical without being the nearest neighbor. Alanine-scan $\Delta\Delta G$ is a cleaner, energetic definition of importance. Separately, since Westhof et al. [1] it has been known that antigenicity correlates with *segmental mobility* (flexibility / B-factor) — an antigen-intrinsic, antibody-agnostic property. The opportunity is to test, with modern MD, whether antigen-only *dynamical* and *solvation* features (RMSF, collective-mode participation, hydration thermodynamics) predict energetic binding importance — and whether they beat static exposure. This is where the author's methods background (FRESEAN collective-mode analysis; 3D-2PT-style hydration thermodynamics; non-equilibrium energy flow) is the differentiator (`../../_literature/my_manuscripts/`).

2 Design decisions to hold onto

Five issues decide whether the project produces a result or an artifact. Recorded here so they stay front-of-mind.

1. **The label is antibody-specific; the features are antibody-agnostic.** $\Delta\Delta G^{(i)}_{\text{bind}}$ is a residue's contribution to *one* paratope, but $\mathbf{x}_i$ comes from the apo antigen and cannot know

the antibody. So $f(\mathbf{x}_i) \to \Delta\Delta G^{(i)}$ can only recover the *antibody-marginal* (antigen-intrinsic) hotspot propensity. This is not a weakness to hide — it is exactly why dynamics/flexibility features should help (cf. Westhof). Where an antigen has several antibodies, average $\Delta\Delta G^{(i)}$ over them to *define* the intrinsic target.

2. **"Beyond SASA" is the primary hypothesis, not a footnote.** Surface exposure is the master confounder: exposed residues are simultaneously more likely to be hotspots, more flexible, and more hydrated. The publishable claim is not "flexibility predicts epitopes" (Westhof has that) but "dynamics/solvation features add signal *after* controlling for static exposure." Design: baseline = SASA + depth + residue type; then test the partial R^2 added by RMSF, mode participation, and 3D-2PT hydration.

3. **Effective $N \approx$ #antigens/folds, not #residues.** Residues within an antigen are correlated and the intrinsic signal is per-fold. Split **leave-one-complex-out** (ideally leave-one-fold-out); never random residue splits. At this scale the work is hypothesis-generating — powered for a strong dynamics-vs-exposure effect, not a deployable predictor. State this plainly.

4. **Features come from the apo antigen** — which is the point and sets a ceiling. Cryptic / induced-fit epitopes (conformational selection vs. induced fit) cannot be predicted in principle from apo features. Conformational-selection epitopes *should* show in apo flexibility/mode features; pure induced-fit ones will not. That contrast is itself a result.

5. **The MD protocol is not one stack.** A flexibility/RMSF run differs from a FRESEAN/3D-2PT run (the latter needs high-frequency velocity output, constraint removal, and a water model validated for dynamics). Likely two protocols per antigen. The house TIP3P/2 fs/LINCS default is scrutinized accordingly (see `../docs/METHODS.md`).

3 The label benchmark

Curated experimental alanine-scan $\Delta\Delta G$ for antigen-side residues lives in `benchmark/antigen_alanine_scanning` (6 sheets: README, Simple_Primary, Full_Schema [51 cols], hGH_Table2_4, Paper_Inventory, Counts). Audited 2026-06-18.

Coverage. 203 primary rows, 13 datasets, 7 antigens. Source: AB-Bind [2] (195 rows; `github.com/sarahsirin/A` and the Jin 1994 hGH thesis (8 rows). Sign convention: $+\Delta\Delta G \Rightarrow$ weaker binding $\Rightarrow$ residue important (38 negative = binding-improving alanines; 17 zeros).

Integrity. The $\Delta\Delta G \leftrightarrow K_d$-ratio arithmetic is internally consistent for all 203 rows ($< 2\%$ vs $\exp(\Delta\Delta G/RT)$). No censoring in the primary set. The 4 non-alanine rows (hGH, EC50-derived) are correctly flagged `exclude_from_binding_label`. Provenance (DOI, source line, URL) is complete except the two homology-modeled BoNT/A2 sets and the Jin rows. `transcription_confidence` is uniformly `high` — treat as a soft flag, possibly a default fill. `[verify]` spot-check 1–2 known hotspots against the source papers.

Clean MD-correlation core. Dropping homology models (`HM_`) and EC50/excluded rows leaves **163 rows / 5 antigens on real PDBs** (VEGF, lysozyme, HPr, BoNT/A1, MT-SP1).

4 First system: hen egg-white lysozyme (HEL)

HEL is the cleanest entry point: small (129 residues), rigid (apo ≈ bound), very stable, textbook AMBER behavior, and — crucially — it carries **three** independent antibody alanine scans in the benchmark, on a consistent 1–129 numbering:

Antibody	PDB	Ag chain	# Ala	$\Delta\Delta G$ range (kcal/mol)
HyHEL-63	1DQJ	C	11	0.60–6.10
D1.3	1VFB	C	12	0.20–2.90
HyHEL-10	3HFM	Y	4	1.03–6.83

Why this is better than one antibody. HyHEL-10 and HyHEL-63 share an epitope and *agree* on hotspots (K96: 6.83 vs 6.10; Y20: 4.87 vs 3.30; K97: 6.18 vs 3.50; R21: 1.03 vs 1.30) — direct evidence for the antigen-intrinsic hotspot idea (Design note 1). D1.3 binds a *non-overlapping* epitope (residues 18–24 and 116–129; zero residue overlap with the HyHELs), giving a second epitope plus built-in negative controls. Because we extract features from the apo antigen, **one apo-HEL trajectory** serves all three antibodies and their mean.

Starting structure. RCSB **1AKI** (1.5 Å X-ray, chain A, apo). Verified: the 1AKI sequence matches the benchmark WT identity at all 23 labeled positions, so labels map 1:1 onto 1–129 numbering (K96/K97/Y20 hotspots confirmed). Apo crystal chosen over antigen-stripped-from-complex to match the prediction scenario; HEL's rigidity makes the choice low-risk.

`figures/hel_cr`
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5 Methods — apo-HEL run #1

Frozen, signed-off stack in `../docs/METHODS.md` (2026-06-18). Purpose of run #1: *validate the prep→equil→production pipeline on Gemini and extract a first RMSF / flexibility feature* (the Westhof re-test). House default stack signed off as the starting point.

Stack. AMBER99SB-ILDN / TIP3P; dodecahedron box, 1.2 nm clearance; neutralize + 0.15 M NaCl; leapfrog, 2 fs, LINCS on h-bonds; PME ($r_{\text{coul}} = r_{\text{vdw}} = 1.0$ nm, potential-shift, DispCorr); V-rescale at 300 K (Protein / Non-Protein); EM → 100 ps NVT → 100 ps NPT (both position-restrained) → 20 ns production. Explicit `gen_seed` and `ld_seed` for reproducibility. Frozen `.mdp` files: `../md/apo_hel/configs/`.

Deliberate deviations from the house scripts. (i) **C-rescale** barostat for NPT equilibration (Parrinello-Rahman only in production), since PR can ring when started far from equilibrium; (ii) reconciled the box type the old scripts disagreed on (dodecahedron); (iii) explicit random seeds.

HEL-specific checkpoints (not glossed over). pdb2gmx must form HEL's **4 disulfides** (6–127, 30–115, 64–80, 76–94) — verify the log reports 4 SS bonds. His15 protonation: record the assigned state. Crystal waters (78 HOH) removed; fresh TIP3P added. Read every grompp warning rather than letting `-maxwarn` hide it.

Caveats / flags. TIP3P over-mobilizes water and 2 fs+LINCS suppress H dynamics — fine for relative RMSF and pipeline validation, *not* for the dynamics-grade features. 20 ns / single replica is a validation length; a converged RMSF feature will need longer + $\geq$ 3 replicas.

Deferred: the distinctive run. FRESEAN mode participation and 3D-2PT hydration thermodynamics require a *separate* protocol — high-frequency velocity output, constraint removal / sub-fs dt, TIP4P/2005 — with settings pulled from the FRESEAN and energy-transfer manuscripts. This is the run that powers the mode-based and hydration feature classes.

6 Run log

Exact-command reproducibility ledger. Append a row per stage; pin the GROMACS version actually used on Gemini, seeds, and Slurm job id.

Date	Stage	Command	GROMACS ver.	Job / outp
2026-06-18	structure	`curl .../1AKI.pdb`	—	`structure`
pending	prep	`sbatch md/apo_hel/prep.sh structures/1AKI.pdb`	`[verify]`	npt.gro/cp
pending	production	`sbatch md/apo_hel/md.sh`	`[verify]`	md.xtc

Status (2026-06-18). Project scaffolded; benchmark audited; first system (HEL/1AKI) selected and label mapping verified 1:1; apo-HEL run #1 staged and signed off. **Next:** confirm Gemini's GROMACS module/version and partition, sync the repo to the Gemini folder, then `sbatch prep.sh`. Open: production length / replicas for a converged RMSF feature; scheduling the FRESEAN/3D-2PT run.

References

[1] E. Westhof, D. Altschuh, D. Moras, A. C. Bloomer, A. Mondragon, A. Klug, and M. H. V. Van Regenmortel. Correlation between segmental mobility and the location of antigenic determinants in proteins. *Nature*, 311:123–126, 1984.

[2] Sarah Sirin, James R. Apgar, Eric M. Bennett, and Amy E. Keating. Ab-bind: Antibody binding mutational database for computational affinity predictions. *Protein Science*, 25(2):393–409, 2016.